AXEOM STUDIO: SYSTEMS ENGINEERING & RECOVERY

Executive Operational Audit, Forensic Reconciliation & 6-Pillar Architecture

PRESENTER & PROJECT IDENTITY

- **Presenter:** Muhammad Abu Huraira (Computer Science Graduate)
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- **Challenge Pack:** Pack 175 (Creative Video & Podcast Studio, Dallas)
- **Business 'Today':** Tuesday, October 6, 2026
- **Live Web Application:** https://axeom-challenge-five.vercel.app
- **GitHub Repository:** https://github.com/hurairamuzammal/axeom-challenge-pack-175

STUDIO FINANCIAL RADAR AT A GLANCE

- **Total Invoiced Revenue:** USD 103,890.00 (138 Invoices Across 2026)
- **Cash Collected in Bank:** USD 83,390.00 (122 Reconciled Bank Deposits)
- **Delinquent Overdue Debt:** USD 15,850.00 (8 Invoices Past Net Terms)
- **Trapped Creative Work:** USD 8,210.00 (12 Delivered Video Cuts Never Invoiced!)
- **Cash Owed to Clients:** USD 0.00 (Completed Rescheduled Shoot B154; INV-1138 Voided)
- **Operational Diagnosis:** Strong market demand crippled by disconnected data silos

HOW I DIVIDED THE 72 HOURS: AN ENGINEERING SPRINT

Disciplined Phased Execution: From Raw Data Ingestion to Reactive Architecture

DAY 1 (HOURS 00 - 24): FORENSIC INGESTION & IDENTITY

- **Schema & Date Audit:** Inspected all 8 raw files line-by-line; resolved European DD/MM/YYYY dates in payments.csv.
- **Lead Hygiene:** Quarantined 4 bot/crypto spam leads (.ru / Telegram) into audit log without losing data.
- **Entity Deduplication:** Collapsed 72 messy surface names into 29 canonical client accounts.
- **Agency Payer Isolation:** Separated third-party payer Fieldhouse Marketing from clinical clients.

DAY 2 (HOURS 24 - 48): FINANCIAL & PRODUCTION RECONCILIATION

- **Multi-Pass Ledger:** Matched 122 bank payments against 138 invoices using deterministic FIFO logic.
- **Revenue Leak Solved:** Uncovered USD 8,210.00 in completed, delivered shoots that were never invoiced.
- **Prepayment Guard:** Proved USD 0.00 cash owed to Blue Bonnet Realty; voided duplicate INV-1138.
- **The Eli Breakdown:** Audited 106 cards with 0 notes on board cards under junior editor Eli.
- **TypeSafe AI Jev:** Integrated Jev (docs.typesafe.ai) to triage 200 messages & 6 audio voice notes.

DAY 3 (HOURS 48 - 72): REACTIVE WEB APP & STRESS TESTING

- **Full-Stack Engine:** Engineered reactive state engine with SQLite persistence & REST API.
- **Responsive UI:** Crafted compact cards with zero wasted void space (250px to 1920px).
- **Operational Stepper:** Built 6-stage lifecycle drawer: Booked -> Editor -> Intake -> Approval -> Billing -> Ledger.
- **Live File Upload:** Engineered drag-and-drop ingestion that recalculates all metrics in 1.1s.
- **Automated Test Suite:** Authored 31 unit & scenario stress tests passing in 0.92s (100% pass rate).

WHAT ARE THE MAJOR ISSUES? ROOT CAUSE DIAGNOSIS

Three Systemic Failures Threatening Axeom's Cash Flow, Reputation, and Retention

1. FINANCIAL CHAOS & LEAKS

- **USD 15,850 Overdue Debt:** 8 Invoices past net terms (Apex Fitness $4.5k, Hillcrest PT $5k withheld, Fieldhouse $5.4k).
- **USD 8,210 Unbilled Work:** 12 delivered video cuts across 7 shoots completed with zero invoices sent!
- **Billing Tool Desync:** 6 invoices marked unpaid were already settled in payments.csv; INV-1110 marked paid with $0 deposit.
- **Duplicate Invoicing:** INV-1138 ($1,500) mistakenly billed to Blue Bonnet for prepaid shoot B154.

2. THE COMMUNICATION BLACK HOLE

- **The 'Eli Breakdown':** Junior editor Eli joined July 3, 2026; logged 0 revision notes across all 106 cards!
- **Inflated Production Time:** Eli averages 6.30h per cut (+48% higher than Rita's 4.26h) with a 65% revision rate.
- **Lost Revision Directives:** 6 Audio Voice Notes and DMs lost in transit before reaching editor cards.
- **Zombie Tasks:** V121 and V152 left stranded 'In revision' for 81 and 73 days despite client approval.

3. IDENTITY FRAGMENTATION & CHURN

- **72 Surface Names:** Staff entered messy aliases across forms, causing severe tracking and billing failures.
- **Imminent Account Churn:** Hillcrest Physical Therapy withholding $5,000 in retainers ($30,000 annual contract at risk).
- **Compliance Violation:** Ignored voice note (Sept 2) and email (Sept 8) demanding full name 'Hillcrest Physical Therapy'.
- **Threat to Leave:** Email on Oct 3: 'We need to talk about the contract this week.'

SYSTEMS ARCHITECTURE

HOW I SOLVED IT: THE IDEAL 6-PILLAR ARCHITECTURE

Permanent Deterministic Safeguards Replacing Fragmented Spreadsheets and Chat

PILLARS 1 & 2: INTAKE & RELATIONAL DATA

- **1. Inbound Spam Gatekeeper:** Filters bot scrapers and .ru domains at intake; whitelists verified boutique clients.
- **2. Relational Primary IDs:** Enforces immutable keys: lead_id -> booking_id -> video_id -> invoice_no -> payment_id.
- **Data Integrity:** Permanently prevents orphaned payments, unlinked shoots, and lost billable hours.

PILLARS 3 & 4: AUTOMATED INTAKE & CLIENT MASTER

- **3. WhatsApp/Insta Revision Bots:** Prompts clients for Invoice No or Video ID; TypeSafe AI Jev transcribes voice notes.
- **Direct Board Sync:** Injects revision notes directly onto assigned editor's board card in real time.
- **4. Single-Source Dropdowns:** Mandates picking clients from validated dropdowns at every pipeline stage.
- **Zero Typos:** Stops errors (torresdental.con) and collapses 72 surface variants to 29 canonical accounts.

PILLARS 5 & 6: BALANCE PRE-CHECK & AUTO DUNNING

- **5. Balance Pre-Check Guard:** Verifies prepaid deposits before billing; voids duplicate invoices (Blue Bonnet $0 owed).
- **6. Smart Payment Reminders:** Triggers polite notices 3d before due date, with 7d/14d/30d aging escalation schedules.
- **Suggest, Never Act Alone:** Every reminder draft requires owner one-click approval before sending.

LIVE APPLICATION WALKAROUND & ACTION ROADMAP

Now Jumping to Live Demo at https://axeom-challenge-five.vercel.app

WHAT TO EXPLORE IN THE LIVE APP DEMO

- **1. 6 Top Hero Metric Tiles:** Instant high-level radar: 198 Videos, 24 Clients, $15.8k Overdue, $8.2k Never Billed, $0 Owed.
- **2. Compact Responsive Grid:** Ultra-compact video cards fitting 250px mobile to 1920px ultrawide without wasted void space.
- **3. Amber Action Badges:** Pulsing 'Action Needed: Unbilled ($900)' tags on 12 unbilled delivered video cuts.
- **4. 6-Stage Operations Stepper:** Click any video card (e.g. V152) to trace Booked -> Editor -> Intake -> Approval -> Billing -> Ledger.
- **5. Retainer Contract Insights:** Advance billing schedule (1st of month) with verified total paid-to-date figures.
- **6. Dynamic File Re-Upload:** Click 'Add new files' to drop in CSVs and watch the engine recalculate live in 1.1s.

MONDAY MORNING TOP 3 ACTIONS (EXACT PROOF)

- **Action 1: Save Hillcrest PT ($30k ARR):** Call Dr. Alana Ibarra & Leah Reed at 9:00 AM (Receipts: line 189 email, Lead L097, INV-1116/1144). Assign senior editor Ali to re-render compliant title cards within 24h.
- **Action 2: Bill $8,210 & Void INV-1138:** Issue invoices for 7 completed shoots (Receipts: B020, B024, B080, B038, B083, B131, B132). Link prepaid wire P5121 to completed B154, void duplicate INV-1138 ($0 cash owed).
- **Action 3: Reply to Leads & Clear Zombies:** Reply to 15 warm human leads (Receipts: L087 Mia Torres, L097 Leah Reed, L086 Victor Vega). Move zombie tasks V121 and V152 to Delivered, unblocking billing settlement.